Z + jets in Summer22: answers to the questions of 8 November

Muon channel vs electrons, other b/c taggers, pileup in ee vs μμ, MET range, zoomed plots, legend order

8 November 2026

The questions and short answers

question	answer	pages
Small discrepancy in the muon channel	Muon trigger SF corrected. ee and μμ now agree: data/MC 0.949 and 0.943.	4–10
	No muon choice (pT, η, SFs, scale/smearing) changes the pT shape.	
Other taggers, Loose/Medium/Tight	DeepJet and RobustParT added, each with its own BTV SFs.	20–24
Loose tau ID veto	Not done yet (postponed, as agreed).	–
Why ee and μμ pileup look different	The 20 test files of each channel cover different runs.	11–13
MC colours ee vs μμ	Legend order is now the same in both channels.	14
MET up to ~200 GeV	Done, without the MET cut.	15
Zoom into high-statistics bins	Zoomed copies of 10 plots in every region (linear y, ratio 0.8–1.2).	16–19

Same files and lumisections as the 237-page deck, so everything here compares directly.

Check: all 237 Z→ee histograms are identical to the deck, and the old muon setup reproduces the old Z→μμ numbers exactly (data 65259, MC 67588.4).

How this was run and checked

One run with everything in it

Summer22, the same 20 files per MC sample and per data channel as the 237-page deck. Three taggers × L/M/T × b/c = 18 tagged regions, each with its own BTV SFs. For the muon study, 11 muon setups are filled side by side in the same run, each changing one thing.

Checks before looking at the physics

Z→ee: all 237 histograms bit-for-bit identical to the deck (nothing on the electron side changed). Old muon setup: reproduces the deck's Z→μμ yields exactly. Every sample read all its files.

One file read through a different server

One DY file could not be read from the FNAL disk pool (timeout every time), so it was read through the CMS redirector instead: same file, same events. One EGamma file had a one-off timeout and was rerun.

Muons: what changed, following the MUO recommendations

Trigger SF now uses both muons

IsoMu24 fires if either muon fires, so the event SF is $[1 - (1-\varepsilon^1\text{data})(1-\varepsilon^2\text{data})] / [1 - (1-\varepsilon^1\text{MC})(1-\varepsilon^2\text{MC})]$, with the MUO IsoMu24 efficiencies (muon_Z.json). Before, we used the SF of the leading muon only. That is only right if we require the leading muon to match the trigger object, which we did not.

Both muons pT > 26 GeV (was 25 for the second muon)

The MUO trigger efficiencies start at 26 GeV. $|\eta| < 2.4$ was already used. Leading muon stays > 35 GeV.

Already in, unchanged

Muon scale and smearing (the Run 3 replacement of Rochester), Tight ID and Tight PF isolation SFs, pileup weights, golden JSON, MET filters.

Effect on Z→μμ: data/MC 0.966 → 0.943. Z→ee (0.949) is unchanged.

Most of it is the trigger SF formula. The 26 GeV cut alone changes data/MC by +0.3%.

Muon setups after the full selection

MET < 50 GeV and ≥ 1 jet. Each setup changes one thing w.r.t. the new default.

setup	what changes	data	MC	data/MC	vs new	slope pT(μμ)
nominal	**new default**	**62865**	**66632**	**0.943 ± 0.009**	**–**	**-0.9 ± 1.2**
prev	old setup (deck of 5 Nov)	65259	67588	0.966 ± 0.009	+2.34%	-0.8 ± 1.2
trigLead	trigger SF of leading muon only	62865	64847	0.969 ± 0.009	+2.75%	-1.0 ± 1.2
sub25	muon pT > 25 instead of 26	65259	69373	0.941 ± 0.009	-0.29%	-0.8 ± 1.2
lead26	leading muon pT > 26 (not 35)	64178	68028	0.943 ± 0.009	-0.01%	-0.9 ± 1.2
eta21	|η| < 2.1	55087	58383	0.944 ± 0.009	+0.01%	-0.8 ± 1.3
trigMatch	lead muon matched to trigger, its SF	57328	61260	0.936 ± 0.009	-0.81%	-0.3 ± 1.3
noSS	no muon scale/smearing	62867	66794	0.941 ± 0.009	-0.24%	-1.0 ± 1.2
noTrigSF	no trigger SF	62865	67027	0.938 ± 0.009	-0.59%	-0.8 ± 1.2
noIDIsoSF	no ID + iso SFs	62865	69439	0.905 ± 0.009	-4.04%	-0.4 ± 1.2
noPU	no pileup weights	62865	65639	0.958 ± 0.008	+1.51%	-0.6 ± 1.2

Only the SFs move the normalisation. No choice changes the pT shape.

Slopes (% per 100 GeV, pT(μμ) 0–400 GeV) range from -1.0 to -0.3; the setups share most events, so differences are much smaller than the quoted errors. The two POG trigger SF methods (both muons, or trigger-matched leading muon) agree within 0.8%.

Muon setups before the jet and MET cuts

Trigger, two muons, opposite charge, 71 < m(μμ) < 111 GeV.

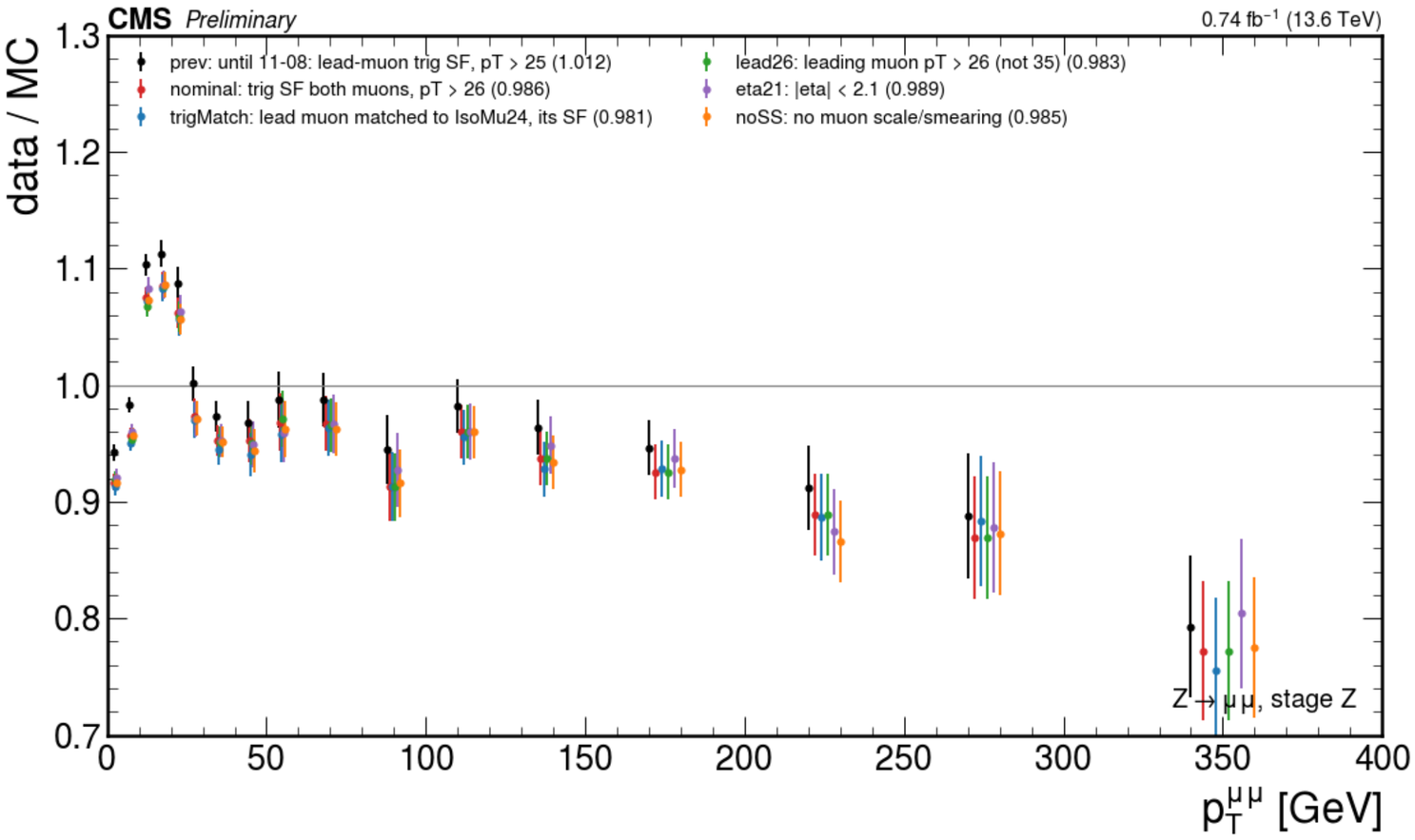
setup	what changes	data	MC	data/MC	vs new	slope pT(μμ)
nominal	**new default**	**387029**	**392392**	**0.986 ± 0.003**	**–**	**-3.5 ± 0.8**
prev	old setup (deck of 5 Nov)	393472	388798	1.012 ± 0.003	+2.60%	-3.8 ± 0.8
trigLead	trigger SF of leading muon only	387029	381794	1.014 ± 0.003	+2.78%	-3.6 ± 0.8
sub25	muon pT > 25 instead of 26	393472	399396	0.985 ± 0.003	-0.12%	-3.6 ± 0.8
lead26	leading muon pT > 26 (not 35)	418413	425798	0.983 ± 0.003	-0.37%	-3.0 ± 0.8
eta21	|η| < 2.1	335447	339262	0.989 ± 0.004	+0.25%	-3.4 ± 0.8
trigMatch	lead muon matched to trigger, its SF	353029	359763	0.981 ± 0.003	-0.51%	-3.4 ± 0.8
noSS	no muon scale/smearing	386700	392661	0.985 ± 0.003	-0.15%	-3.6 ± 0.8
noTrigSF	no trigger SF	387029	394788	0.980 ± 0.003	-0.61%	-3.4 ± 0.8
noIDIsoSF	no ID + iso SFs	387029	409193	0.946 ± 0.003	-4.11%	-3.0 ± 0.8
noPU	no pileup weights	387029	390858	0.990 ± 0.003	+0.39%	-3.6 ± 0.7

Only the SFs move the normalisation. No choice changes the pT shape.

Slopes (% per 100 GeV, pT(μμ) 0–400 GeV) range from -3.8 to -3.0; the setups share most events, so differences are much smaller than the quoted errors. The two POG trigger SF methods (both muons, or trigger-matched leading muon) agree within 0.5%.

Muon setups: data/MC vs pT(μμ), Z selection

Before the jet and MET cuts.

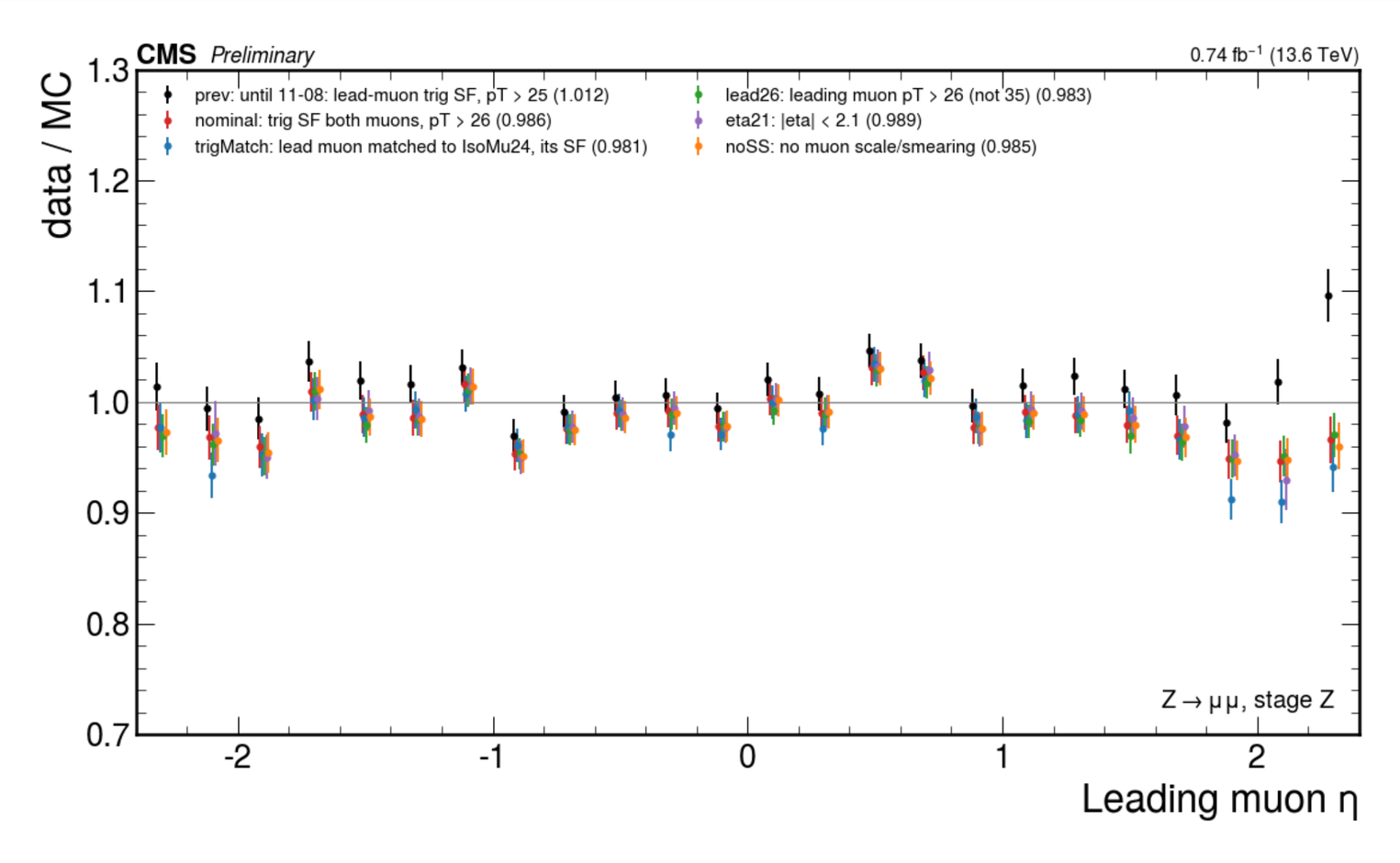


All setups give the same shape; they only shift up or down together.

The fall at high pT(μμ) is there in Z→ee too (next pages), so it is not a muon effect.

Muon setups: data/MC vs leading-muon η

Z selection, before the jet and MET cuts.



Muons with |η| > 2.0 are low: data/MC 0.964 ± 0.010 vs 0.998 in the barrel.

The trigger-matching method gives 0.940 there, so both POG methods see it. Cutting at |η| < 2.1 changes data/MC by only +0.25% (same selection). Not understood yet.

Muons vs electrons, step by step

data/MC per selection step; slope = straight-line fit of data/MC vs pT(ℓℓ), 0–400 GeV, % per 100 GeV.

step	data/MC Z→ee	data/MC Z→μμ	slope Z→ee	slope Z→μμ
all Z events (no jet or MET cut)	0.995 ± 0.005	0.986 ± 0.003	-2.1 ± 1.2	-3.5 ± 0.8
MET < 50, no jet	1.042 ± 0.006	1.030 ± 0.004	+34.3 ± 4.4	+27.1 ± 2.9
MET > 50, no jet	0.416 ± 0.013	0.414 ± 0.008	+61.7 ± 11.3	+48.8 ± 7.0
MET > 50, ≥ 1 jet	0.692 ± 0.027	0.679 ± 0.019	+10.8 ± 4.5	+5.2 ± 2.8
final: MET < 50, ≥ 1 jet	**0.949 ± 0.012**	**0.943 ± 0.009**	**+0.2 ± 1.9**	**-0.9 ± 1.2**

Final selection, data/MC per leading-lepton |η|:

	0–0.8	0.8–1.2	1.2–2.0	2.0–2.5
Z→ee	0.942 ± 0.019	0.909 ± 0.023	0.987 ± 0.032	1.040 ± 0.047
Z→μμ	0.939 ± 0.015	0.915 ± 0.017	0.988 ± 0.020	0.908 ± 0.028

The two channels now behave the same at every step.

The 5–6% deficit after the selection is common: MC has too many events with jets and far too many with MET > 50 GeV.

The pT slope we discussed

Same fit as before (10 GeV bins, final selection, % per 100 GeV) on three runs with the same files.

run	pT(ee)	pT(μμ)	lead μ pT	jet pT (μμ)
29 Oct run, DY Z pT weights ON	+4.8 ± 2.0	+4.4 ± 1.3	+6.2 ± 2.2	+2.0 ± 1.8
237-page deck, weights off	-0.3 ± 1.9	-0.7 ± 1.2	-1.4 ± 2.1	-1.8 ± 1.8
this run (new muon setup)	-0.3 ± 1.9	-0.8 ± 1.2	-1.7 ± 2.1	-2.1 ± 1.8

The +4.4 ± 1.3% per 100 GeV came from the run with the DY Z pT weights on.

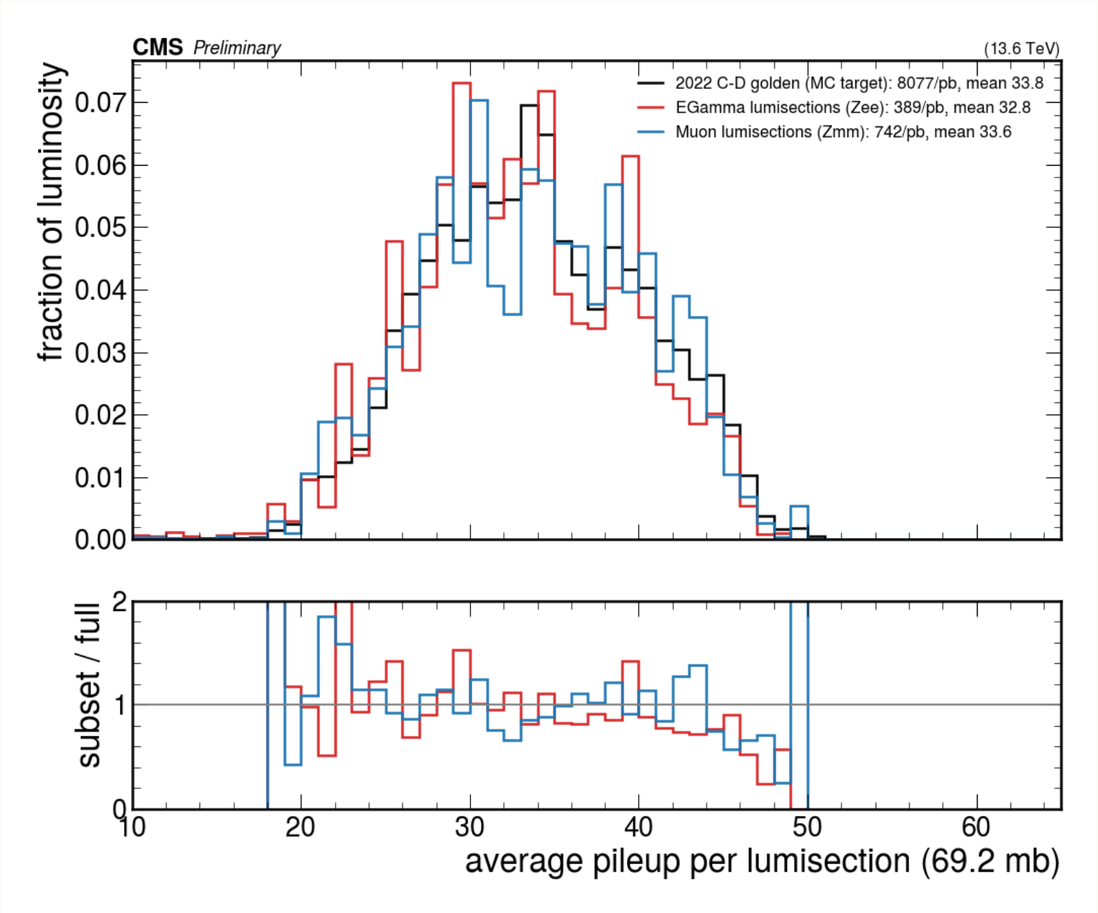
With the weights off (the 237-page deck and now) neither channel has a significant slope after the full selection.

Before the jet and MET cuts both channels fall at high pT($\ell\ell$) in the same way.

Z→ee -2.1 ± 1.2, Z→μμ -3.5 ± 0.8 (% per 100 GeV): the DY generator Z pT, seen before. Still open: a dip at pT($\ell\ell$) 30–60 GeV (≈ 0.92) in both channels.

Pileup: which lumisections the test files cover

Luminosity-weighted average pileup per lumisection (brilcalc, normtag, 69.2 mb).



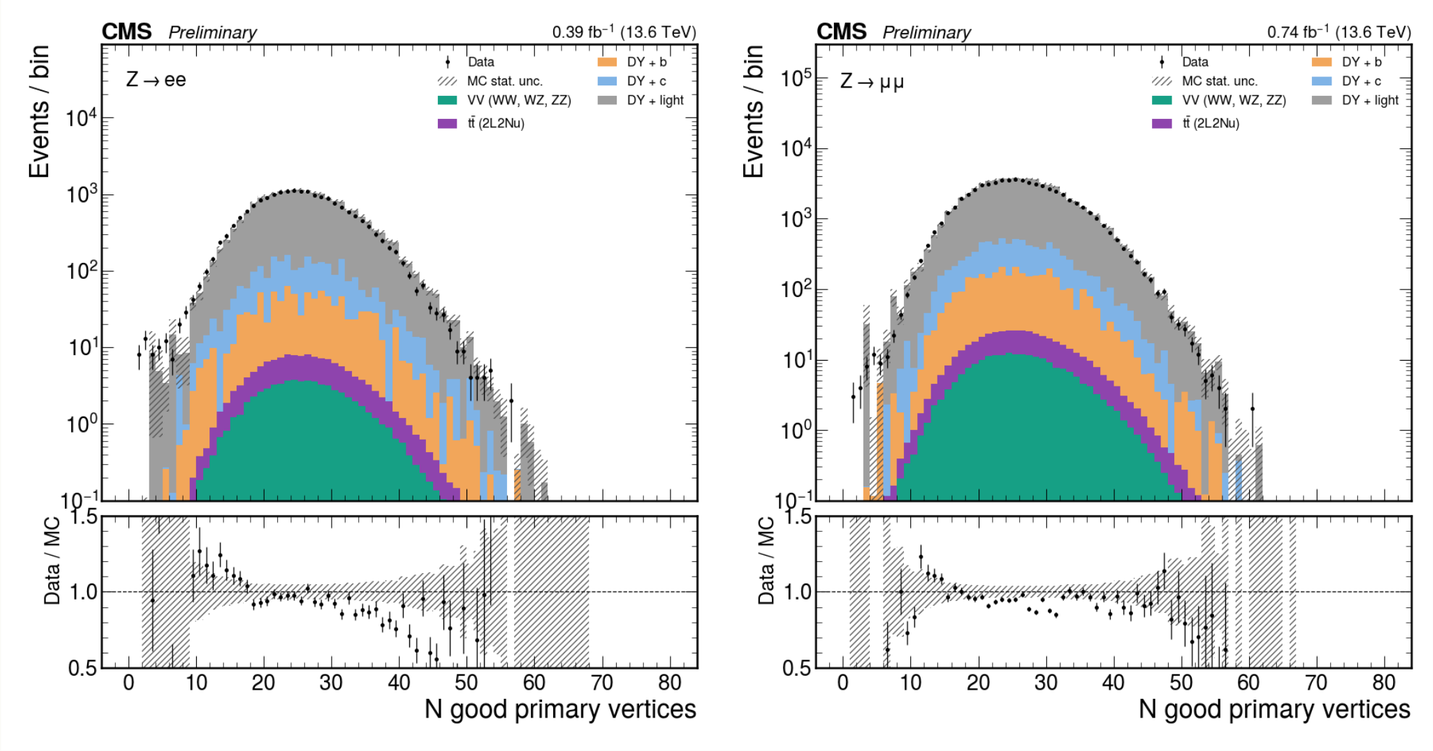
Mean pileup: EGamma files 32.8, Muon files 33.6, full 2022 C–D 33.8.

The MC is weighted to the full 2022 C–D profile, so Z→ee data sit lower in N(PV) than its MC.

Pileup: the ee vs μμ difference is the data sampling

	data − MC, mean N(PV)	pileup of files − full era	expected shift in N(PV)
Z→ee	-0.57	-0.92	-0.64
Z→μμ	-0.14	-0.20	-0.14

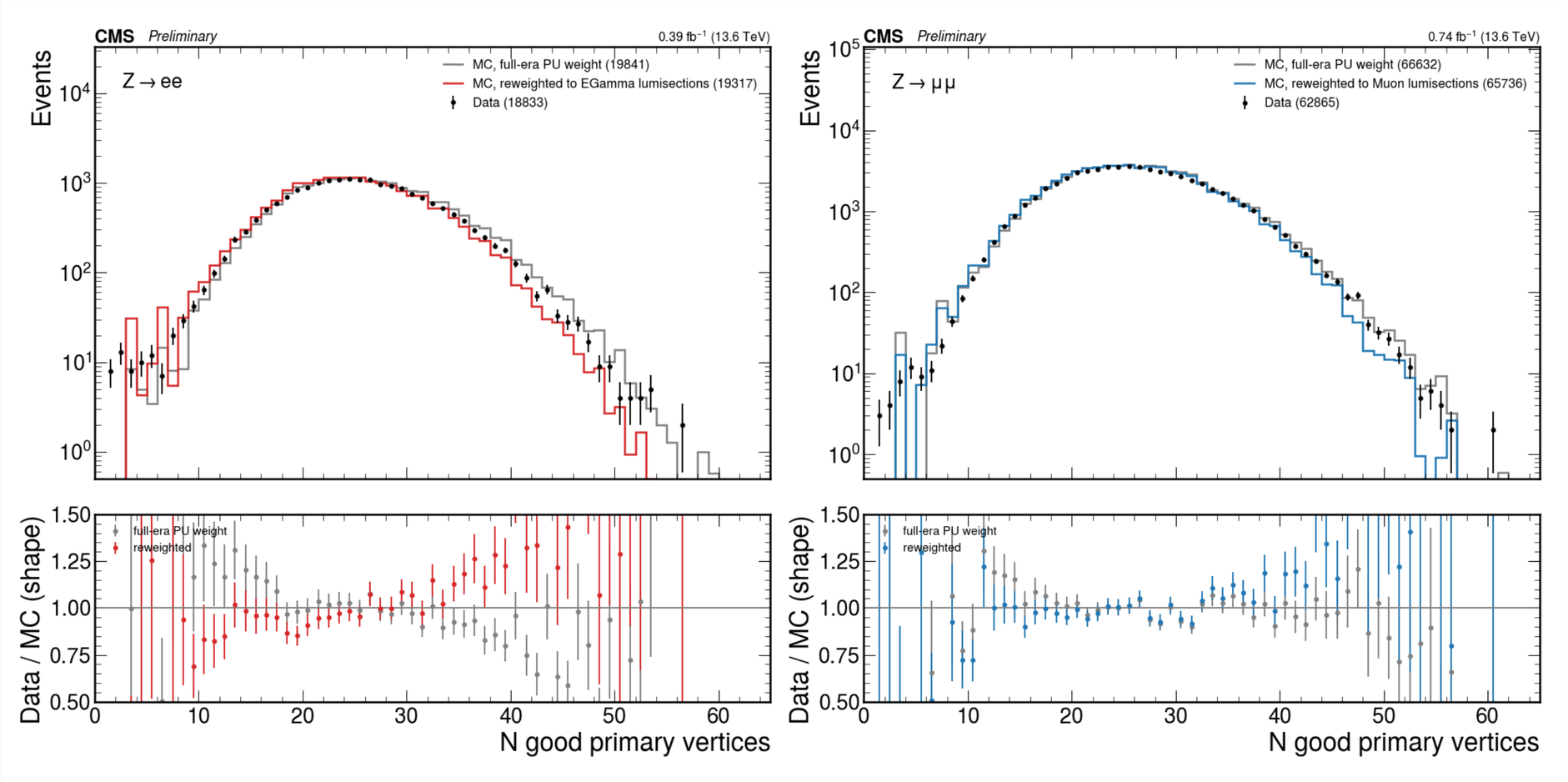
Expected = pileup difference × N(PV) per pileup interaction from the MC (0.69).



The N(PV) offset of each channel is what its lumisections predict.

With the full dataset both channels see the same runs, so this should go away.

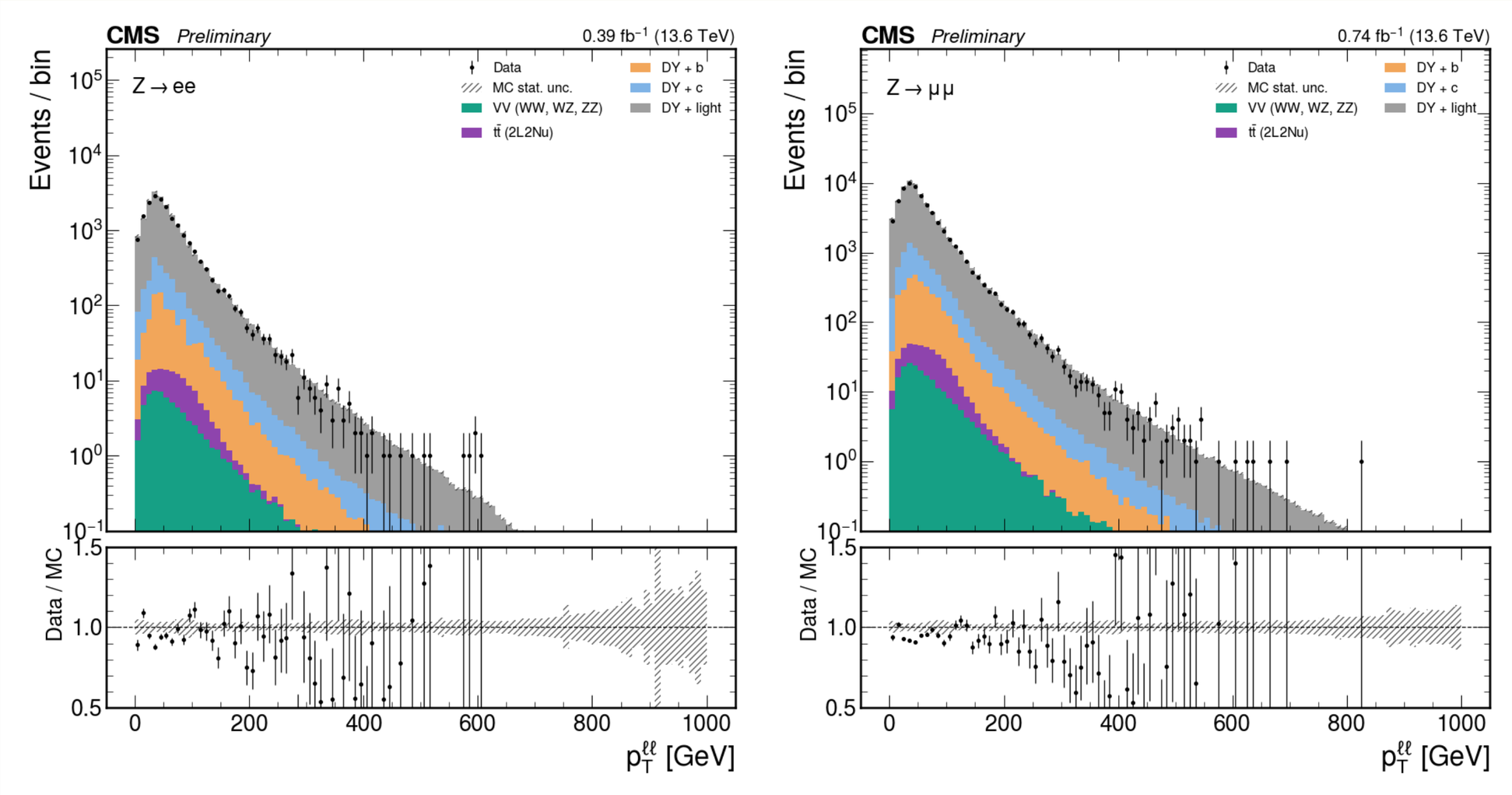
Pileup: a bin-by-bin reweighting overshoots, so it is not used



Reweighting the MC to each channel's own profile over-corrects (Z→μμ, flat before, gets a slope).

Probably because the per-lumisection average ignores the bunch-to-bunch spread. The mean check on the previous page does not depend on that. The clean test is the full-dataset run.

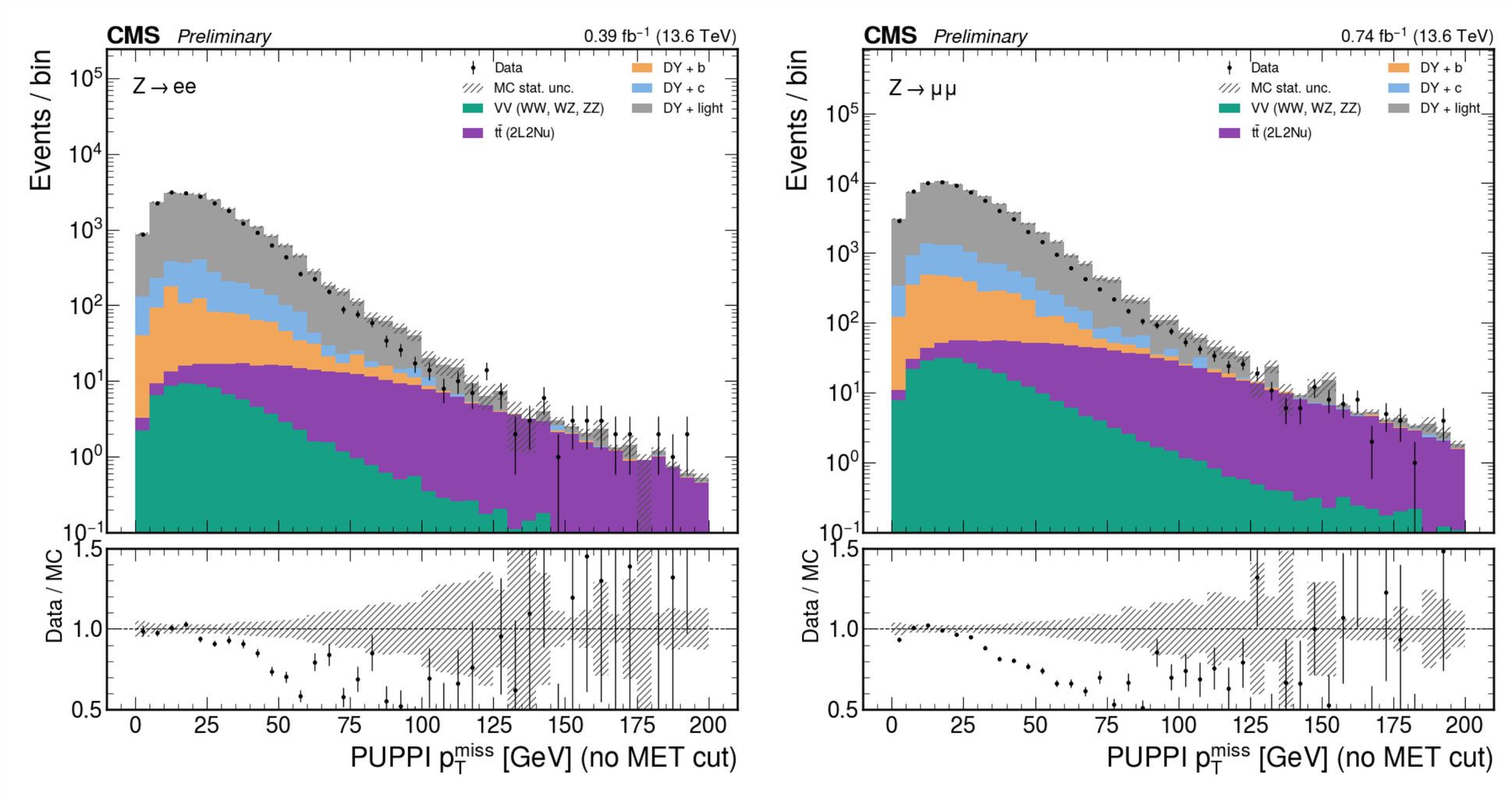
Legend order is now the same in both channels



Before, each channel sorted the stack by its own yields, so DY+b and DY+c swapped places in Z→ee.

Now one order for both: DY+b, DY+c, DY+light, tt, VV. Colours unchanged.

MET up to 200 GeV, without the MET cut

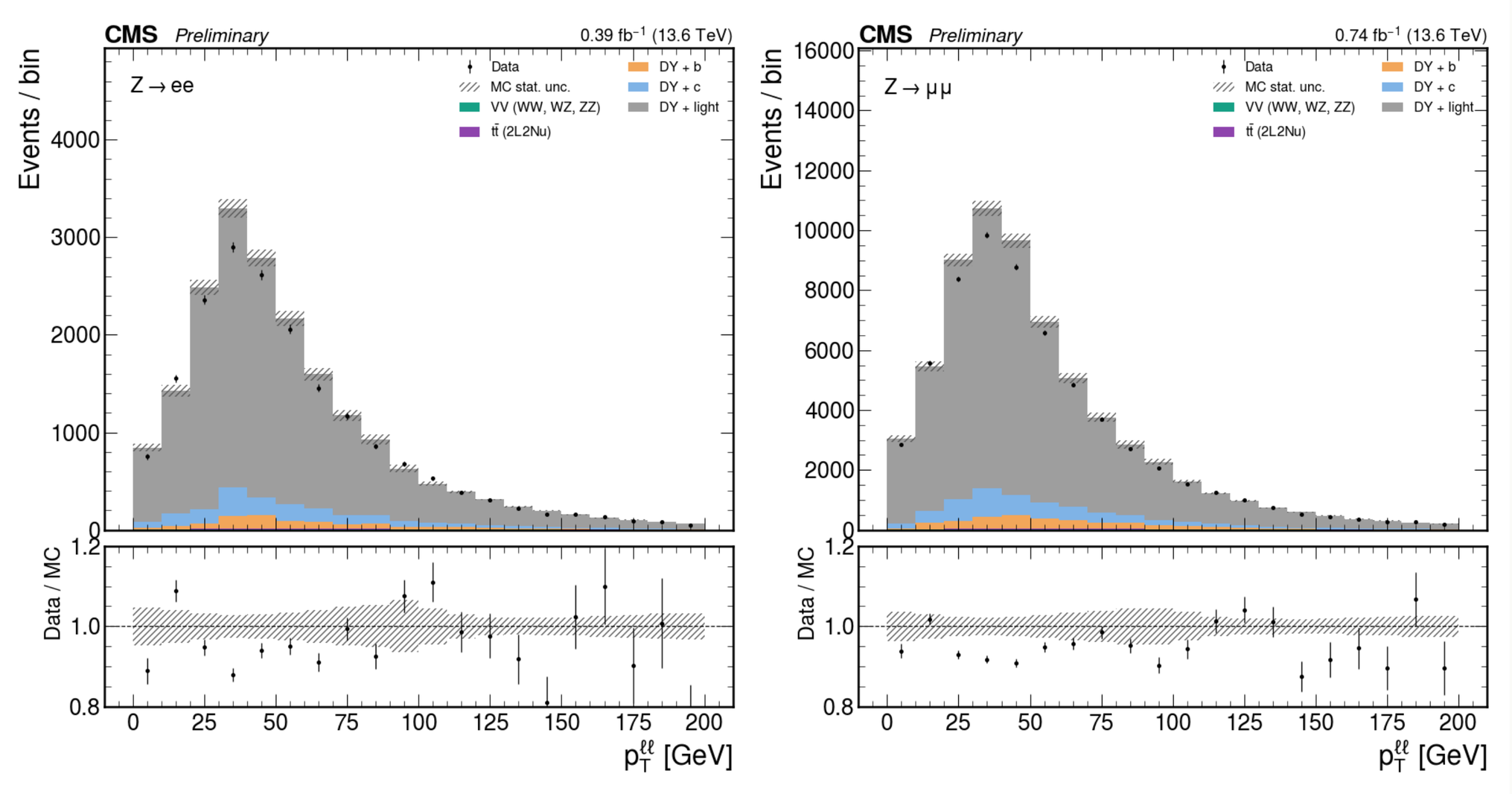


MC is well above data from 40 to 120 GeV in both channels (data/MC down to about 0.6).

This is the 2022 jet/MET issue we traced in October (jets in the type-1 MET). Above ~120 GeV tt dominates.

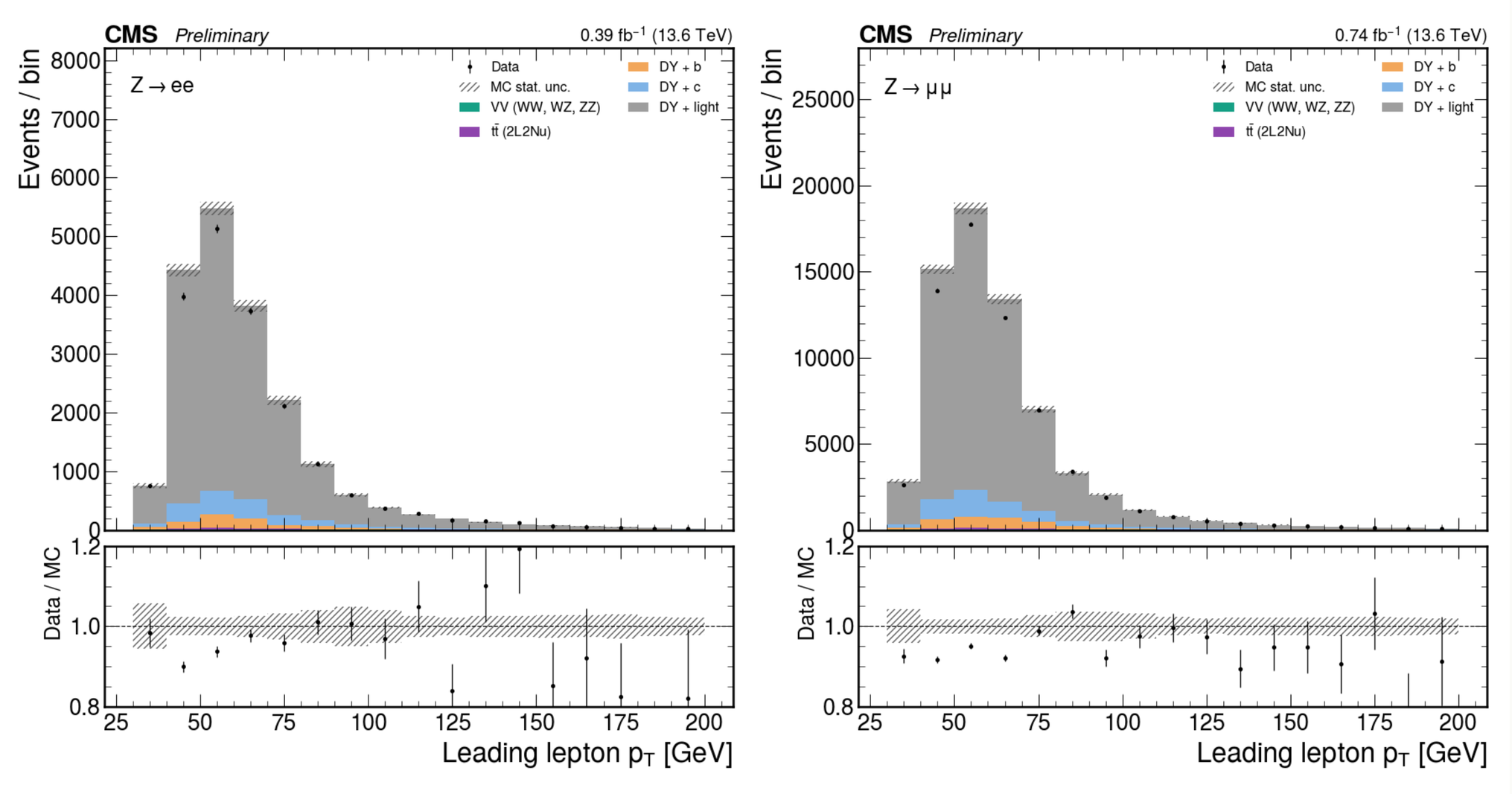
Zoom: Z pT, 0–200 GeV

Linear scale, ratio 0.8–1.2. Every region has zoomed copies of z_pt, lep0_pt, lep1_pt, jet_pt, jet1_pt, lhe_vpt, met_pt_nm1, npv, m_jj, pt_balance (*_zoom.png).



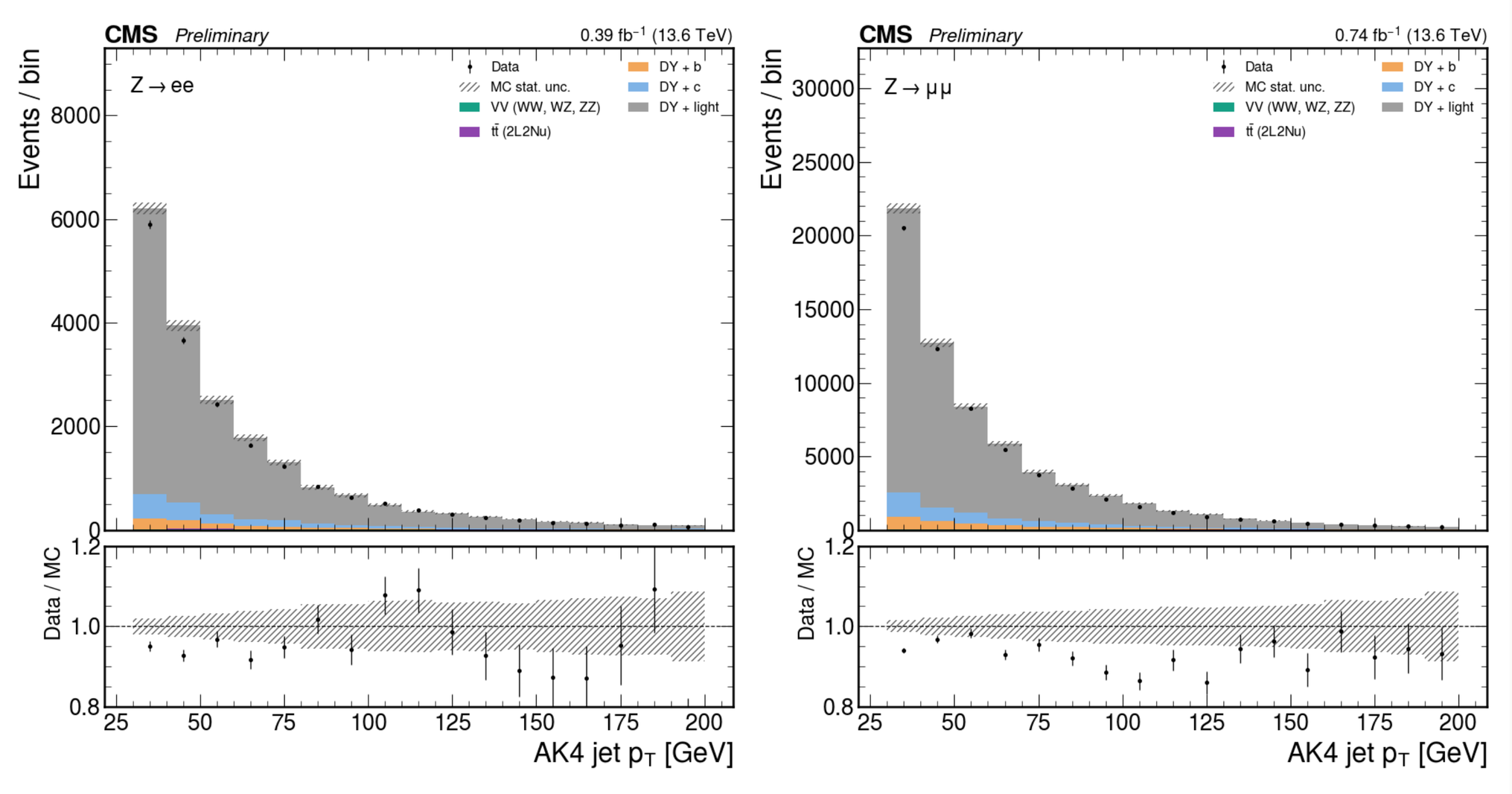
Zoom: Leading-lepton pT, 30–200 GeV

Linear scale, ratio 0.8–1.2. Every region has zoomed copies of z_pt, lep0_pt, lep1_pt, jet_pt, jet1_pt, lhe_vpt, met_pt_nm1, npv, m_jj, pt_balance (*_zoom.png).



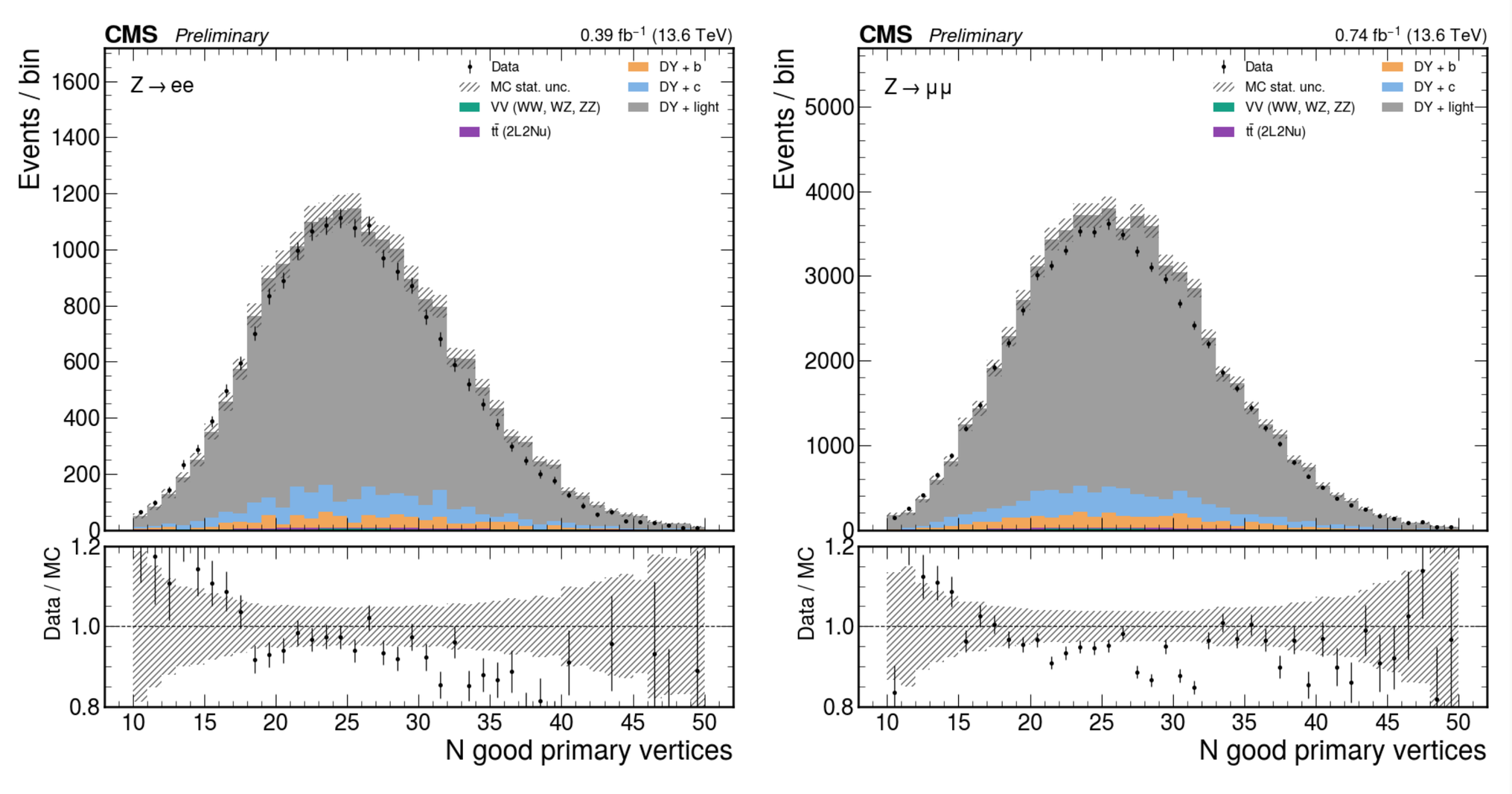
Zoom: Leading-jet pT, 30–200 GeV

Linear scale, ratio 0.8–1.2. Every region has zoomed copies of z_pt, lep0_pt, lep1_pt, jet_pt, jet1_pt, lhe_vpt, met_pt_nm1, npv, m_jj, pt_balance (*_zoom.png).



Zoom: N primary vertices, 10–50

Linear scale, ratio 0.8–1.2. Every region has zoomed copies of z_pt, lep0_pt, lep1_pt, jet_pt, jet1_pt, lhe_vpt, met_pt_nm1, npv, m_jj, pt_balance (*_zoom.png).



Three taggers, Loose / Medium / Tight

Cut on the leading jet. Working points and SFs from the BTV Summer22 files; each tagger uses its own SFs.

tagger	WP	b-tag: B >	c-tag: CvL >	and CvB >
PNet	Loose	0.047	0.054	0.181
	Medium	0.245	0.16	0.306
	Tight	0.6734	0.492	0.259
DeepJet	Loose	0.0583	0.042	0.208
	Medium	0.3086	0.108	0.299
	Tight	0.7183	0.303	0.243
RobustParT	Loose	0.0849	0.039	0.068
	Medium	0.4319	0.117	0.13
	Tight	0.8482	0.36	0.095

SFs: _comb (b, c) and _light for b-tag; _wc (c), _tnp (b) and _light for c-tag.

Same SF validity range for all three taggers. b-tag and c-tag are separate regions.

b-tag: purity and data vs MC per tagger

Purity = DY b-jet share of the tagged DY MC. Tag ratio = tagged fraction in data / in MC: the overall 5% normalisation cancels, so it tests the tagging and its SFs.

tagger	WP	purity ee	purity μμ	tagged, μμ data	tag ratio Z→ee	tag ratio Z→μμ
PNet	L	21%	22%	17.5%	0.998 ± 0.031	0.980 ± 0.022
	M	58%	62%	5.8%	1.011 ± 0.055	0.969 ± 0.038
	T	89%	88%	3.3%	1.058 ± 0.077	0.925 ± 0.047
DeepJet	L	18%	20%	18.9%	0.976 ± 0.029	0.979 ± 0.022
	M	55%	58%	6.0%	1.056 ± 0.056	0.996 ± 0.038
	T	89%	87%	3.1%	1.050 ± 0.081	0.929 ± 0.048
RobustParT	L	21%	23%	17.9%	1.013 ± 0.031	1.040 ± 0.024
	M	61%	62%	5.7%	1.079 ± 0.061	1.006 ± 0.040
	T	85%	89%	3.2%	1.034 ± 0.074	0.963 ± 0.050

Data and MC tag the same fraction of jets within the stat. errors, for all three taggers.

Largest deviation: RobustParT L, Z→μμ: 1.040 ± 0.024 (1.7σ, stat. only). Mean tag ratio over the 6 numbers: PNet 0.990, DeepJet 0.998, RobustParT 1.022. SF uncertainties not included yet.

c-tag: purity and data vs MC per tagger

Purity = DY c-jet share of the tagged DY MC. Tag ratio = tagged fraction in data / in MC: the overall 5% normalisation cancels, so it tests the tagging and its SFs.

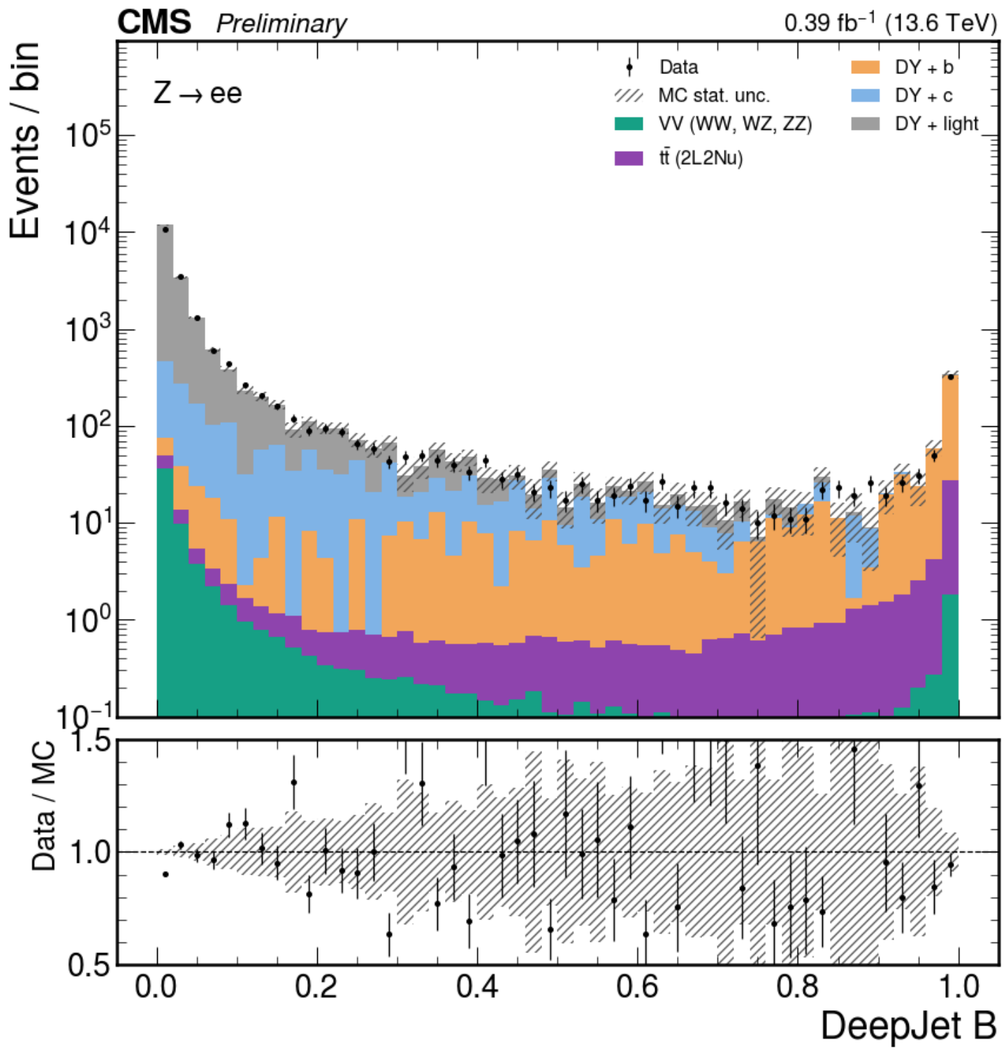
tagger	WP	purity ee	purity μμ	tagged, μμ data	tag ratio Z→ee	tag ratio Z→μμ
PNet	L	9%	9%	88.2%	0.981 ± 0.012	0.985 ± 0.009
	M	20%	20%	28.1%	0.989 ± 0.023	0.985 ± 0.017
	T	50%	51%	7.0%	0.965 ± 0.045	0.963 ± 0.035
DeepJet	L	9%	9%	88.6%	1.015 ± 0.012	1.016 ± 0.009
	M	18%	18%	27.6%	0.994 ± 0.024	1.025 ± 0.018
	T	49%	48%	6.1%	1.088 ± 0.055	1.014 ± 0.038
RobustParT	L	9%	9%	88.4%	1.014 ± 0.012	1.019 ± 0.009
	M	20%	20%	27.5%	1.021 ± 0.024	1.045 ± 0.019
	T	52%	54%	6.6%	1.099 ± 0.056	1.051 ± 0.039

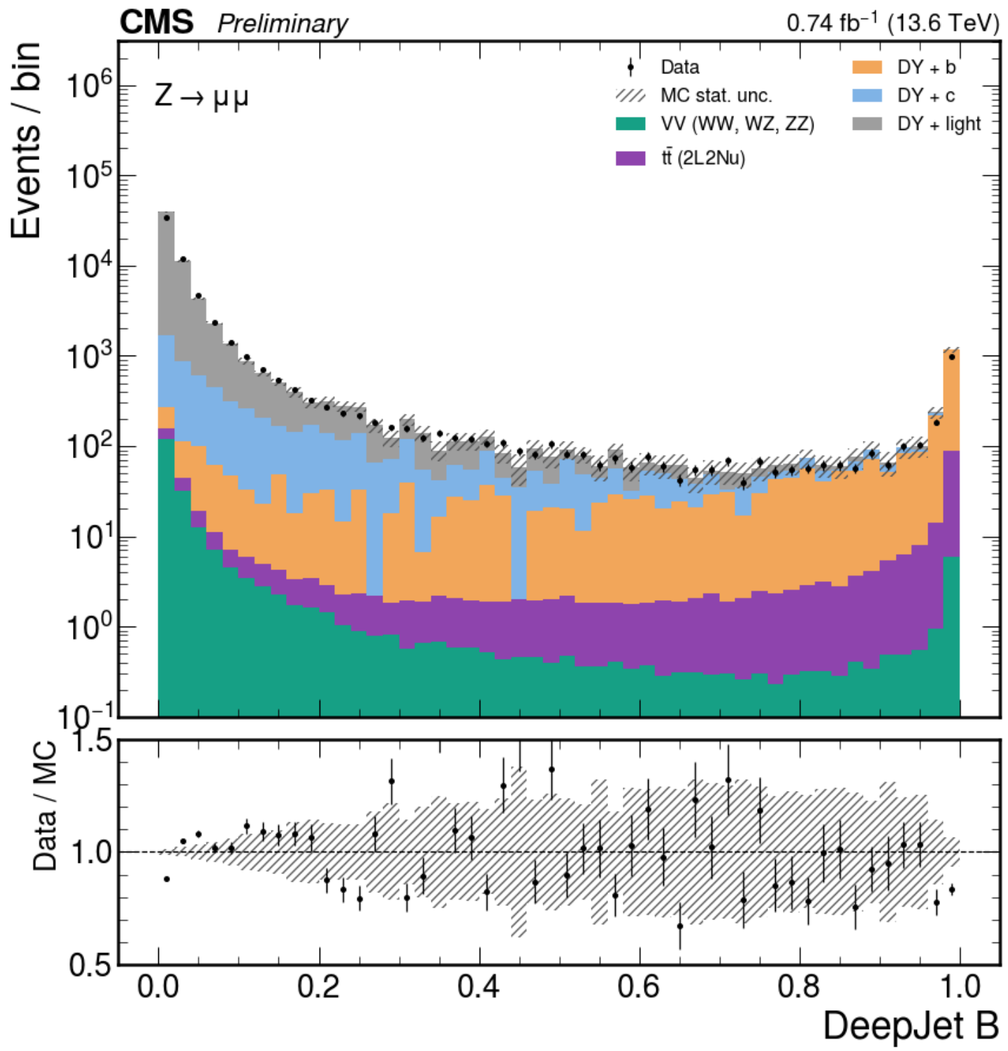
Data and MC tag fractions agree within 10%, but not all within the stat. errors.

Largest deviation: RobustParT M, Z→μμ: 1.045 ± 0.019 (2.4σ, stat. only). Mean tag ratio over the 6 numbers: PNet 0.978, DeepJet 1.025, RobustParT 1.042. SF uncertainties not included yet.

DeepJet B score of the leading jet

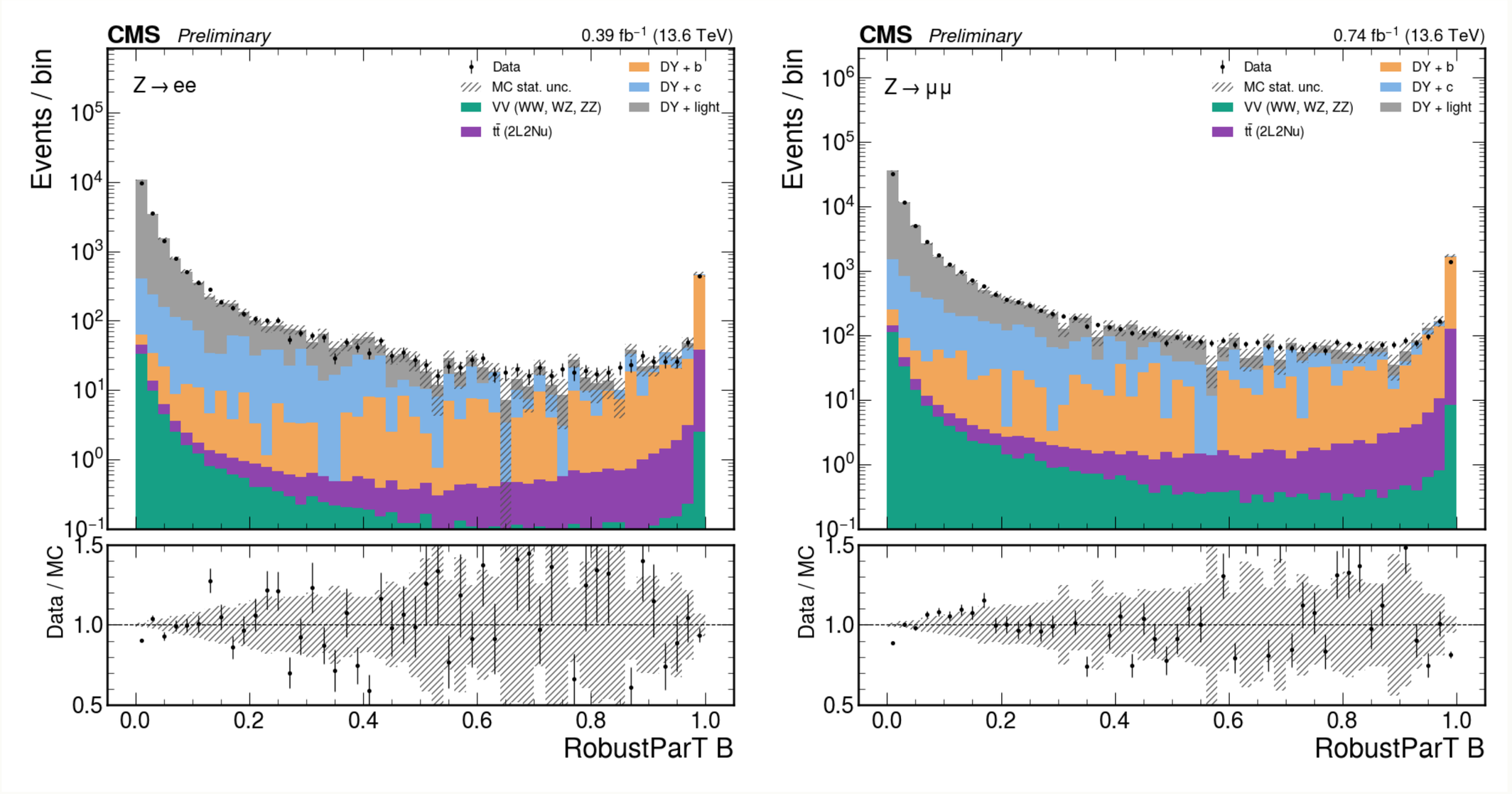
No tag cut, so no tagging SF here (SFs apply to tagged jets only).





RobustParT B score of the leading jet

No tag cut, so no tagging SF here. CvL, CvB plots and all 18 tagged regions: hists_zbzc_run3_v2/Summer22/plots_<region>/.



Summary

Muon channel: trigger SF now follows MUO for two muons. Z→ee 0.949, Z→μμ 0.943.

The channels agree at every step. No muon choice changes the pT shape; there is no significant pT slope after the selection in either channel.

Pileup ee vs μμ: explained by the runs the 20 test files cover.

Legend order fixed, MET up to 200 GeV, zoomed plots in every region.

DeepJet and RobustParT added at L/M/T, with their own SFs. Purity is similar to PNet.

b-tag: data/MC tag ratios consistent with 1 for all three. c-tag: mean tag ratio PNet 0.978, DeepJet 1.025, RobustParT 1.042 (stat. only, no SF uncertainties yet).

Not fixed / still open

• Common 5–6% deficit after the jet and MET cuts: MC has too many events with jets and with MET > 50 GeV.
• Dip at pT($\ell\ell$) 30–60 GeV in both channels. • Muons with |η| > 2.0 about 3% low before the jet cut.
• Loose tau veto not done. • No SF uncertainties yet. • 20 files per sample: stat. errors of 1–8%.

To decide

Keep the new muon setup (pair trigger SF, pT > 26) as the default? Which tagger and WP for b and for c? Then: run the full Summer22 dataset.